

Prediction

Diagnostics summary

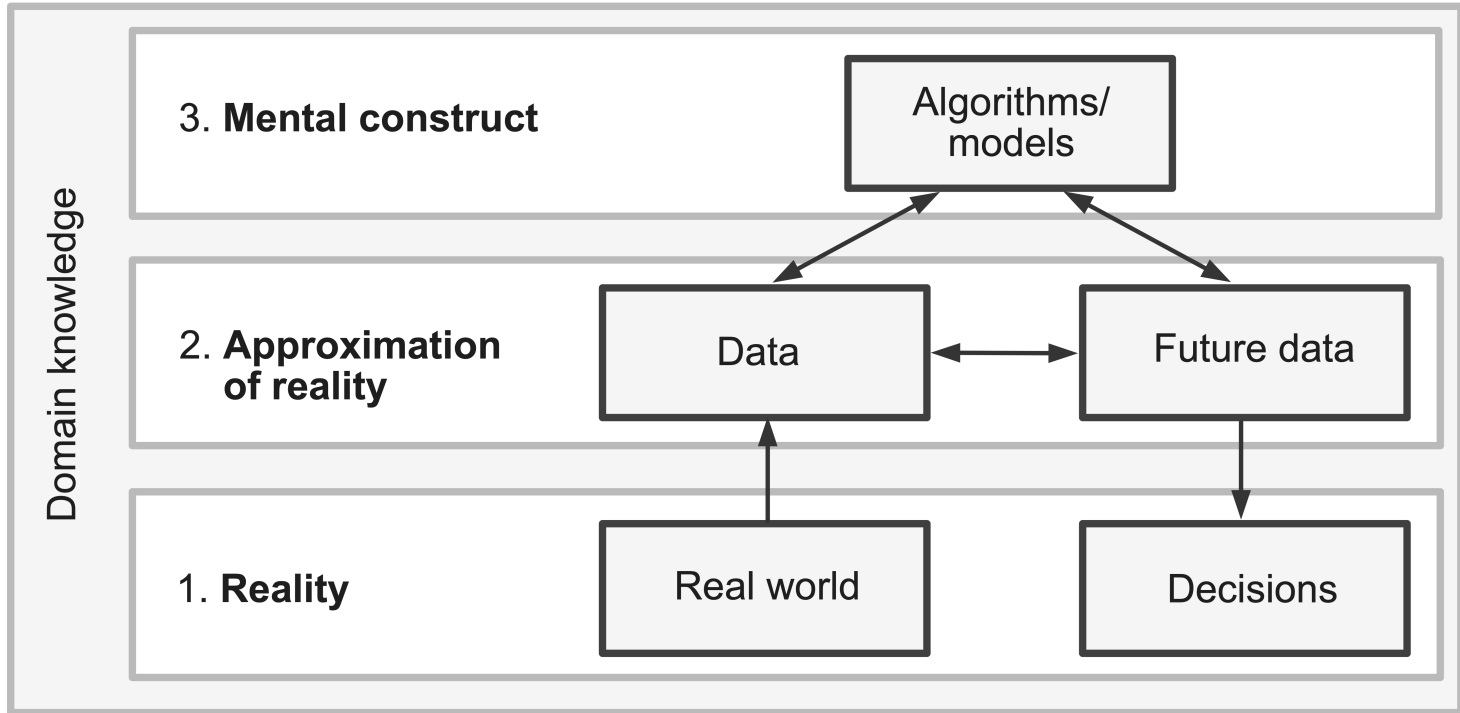
Before fitting a model:

- + Check empirical logit plots; make transformations as needed
- + Investigate any unusual observations
- + Check for multicollinearity with scatterplots / correlations between quantitative explanatory variables; if needed, choose a subset of variables or combine variables

After fitting a model:

- + Check quantile residual plots
- + Check for multicollinearity with VIFs
- + Check for influential points with Cook's distance
 - + If influential points are clearly an error, can remove
 - + Otherwise, report results with and without influential points

Using logistic regression



Is our model good at predicting the response?

Making predictions with the Titanic data

- + For each passenger, we calculate $\hat{\pi}_i$ (estimated probability of survival)
- + But, we want to predict *which* passengers actually survive

How do we turn $\hat{\pi}_i$ into a binary prediction of survival / no survival?

Confusion matrix

```
m1 <- glm(Survived ~ Age + Sex + as.factor(Pclass),  
          data = titanic, family = binomial)  
  
table(Prediction = m1$fitted.values > 0.5,  
      Truth = titanic$Survived)
```

```
##           Truth  
## Prediction    0    1  
##      FALSE 356   83  
##      TRUE  68  207
```

Did we do a good job predicting survival?

Confusion matrix

```
##           Truth
## Prediction    0    1
##      FALSE 356   83
##      TRUE  68  207
```

Class activity

- + Work on the class activity (course website)
- + We will then discuss as a group
- + At the end of class, submit your rendered HTML on Canvas

Class activity

```
m1 <- glm(Default ~ log(DisbursementGross) + Term +  
          as.factor(UrbanRural),  
          family = binomial, data = sba)  
  
table(Prediction = m1$fitted.values > 0.5,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3989  734  
##      TRUE   100  168
```

- + Accuracy =
- + Sensitivity =
- + Specificity =
- + PPV =

Class activity

```
##              Truth
## Prediction FALSE TRUE
##      FALSE  3989  734
##      TRUE   100  168
```

Is an accuracy of around 80% good?

Class activity

Changing thresholds:

```
table(Prediction = m1$fitted.values > 0.3,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  3524  351  
##      TRUE   565  551
```

```
table(Prediction = m1$fitted.values > 0.7,  
      Truth = sba$Default)
```

```
##           Truth  
## Prediction FALSE TRUE  
##      FALSE  4089  902
```